

Testing model-dependent behaviour

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Testing pure components

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Authentication and authorisation

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Architecture and data flow

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Testing model-dependent behaviour. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Testing model-dependent behaviour. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Four runs on each of two cases: first passes 4, second passes 2. Compute overall run rate and per-case rates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A deterministic mocked output tests the pipeline contract but cannot estimate an unspecified live model distribution. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Four runs on each of two cases: first passes 4, second passes 2. Compute overall run rate and per-case rates.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled testing model-dependent behaviour evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of testing model-dependent behaviour. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Testing model-dependent behaviour

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

Testing model-dependent behaviour

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify facts

The answer must retain Friday and room B.



2. Check first reply

Its four words preserve both supplied facts.



3. Check second reply

It is short, but room D contradicts room B.

CHANGE ONE THING

Use a 25-word answer that preserves both facts.

Explain the changed result using the matching lesson.

Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? It is short, but room D contradicts room B.

1. Check second reply
2. Check first reply
3. Identify facts

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check second reply It is short, but room D contradicts room B. A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Practice 2: explain the deciding evidence

Changed-case practice: Use a 25-word answer that preserves both facts. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. It passes support but fails the length requirement.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: It passes support but fails the length requirement. Every mandatory criterion must pass; facts do not erase a length violation.

Repair a missing building block

Review Testing pure components (ai-6/7). Goal: Explain the mechanism and terminology of testing pure components. Short practice: Predict the formatter outputs for 7 and 12, including spacing. Why does input 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the mechanism and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C. Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the mechanism and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
